

Classical vortex solution of the Abelian Higgs model

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An exact vortex solution for the Abelian Higgs model is found when a particular relation between the coupling constants is satisfied. For this case (in which the masses of the scalar and vector bosons are the same), we decouple the classical field equations and solve them. We obtain power-series expansions and asymptotic expressions for the solution. The values of the fields are explicitly calculated for a one-quantum-of-flux solution, and bounds are established for the n -quanta case. Finally, we discuss the field form factors.

I. INTRODUCTION

Recently there has been wide interest in the study of nonlinear field equations. Particlelike solutions were obtained and applied to the problem of quark confinement in hadron physics. In an attempt to construct quantum field theories of hadrons, the important question of quantizing these finite-energy solutions was also extensively investigated.

One of the first classical models was developed by Nielsen and Olesen.¹ They investigated an example of a field theory with a vortex solution by considering an Abelian gauge field coupled to a charged scalar field. They obtained approximate solutions of the equations of motion and related them to the Nambu string in the strong-coupling limit.

The purpose of this paper is to obtain an exact classical vortex-type solution of this model and to study in detail the vector and scalar field properties.

In Sec. II, we describe the model and calculate the energy-momentum tensor of the system. This tensor will turn out to be important to make an ansatz which simplifies considerably the equations of motion (Sec. III) and allows us to obtain an exact solution when a particular relation between coupling constants holds. This, and the determination of the mass spectrum of the theory, is done in Sec. IV.

In Sec. V, we give the explicit one-quantum-of-flux vortex solution. We discuss some properties of the solution and calculate the form factors for the scalar and vector fields in Sec. VI.

In an appendix we deal with the calculation of some bounds for the fields used throughout the paper.

II. THE MODEL

We start from the Lagrangian density of a Higgs model

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} |(\partial_\mu + ieA_\mu)\phi|^2 + c_2 |\phi|^2 - c_4 |\phi|^4, \quad (2.1)$$

where

$$F_{\mu\nu} = \partial_\nu A_\mu - \partial_\mu A_\nu.$$

The corresponding equations of motion are

$$\square^2 A_\mu = \frac{1}{2} ie(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*) - e^2 \phi A_\mu, \quad (2.2)$$

$$\square^2 \phi + 2ieA_\mu \partial^\mu \phi - e^2 A_\mu A^\mu \phi = 2c_2 \phi - 4c_2 \phi^* \phi^2$$

in the $\partial^\mu A_\mu = 0$ gauge.

This model has vortex-type classical solutions as has been put forward by Nielsen and Olesen¹ in analogy with the Ginzburg-Landau theory of superconductivity.² The flux of vortex lines is quantized in units of $2\pi/e$. The solution corresponding to n units of flux has the form

$$\phi(\vec{r}) = e^{-in\varphi} f(\rho) \quad (n=1, 2, 3, \dots) \\ \vec{A}(\vec{r}) = -\hat{e}_\varphi \frac{A(\rho)}{\rho}. \quad (2.3)$$

This solution depends only on the polar coordinates (ρ, φ) and it is independent of z . The appropriate boundary conditions for finite-energy vortex solutions are

$$f(0) = A(0) = 0, \\ \lim_{\rho \rightarrow \infty} f(\rho) = + \left(\frac{c_2}{2c_4} \right)^{1/2}, \quad (2.4) \\ \lim_{\rho \rightarrow \infty} A(\rho) = -\frac{n}{e}.$$

For these cylindrically symmetric solutions, Eqs. (2.2) take the form

$$\frac{d^2 A}{d\rho^2} - \frac{1}{\rho} \frac{dA}{d\rho} - e(n + eA)f^2 = 0, \quad (2.5) \\ \frac{d^2 f}{d\rho^2} + \frac{1}{\rho} \frac{df}{d\rho} - \frac{(n + eA)^2}{\rho^2} f + 2c_2 f - 4c_4 f^3 = 0.$$

As we pointed out in Sec. I, Nielsen and Olesen found approximate solutions for large ρ :

$$A(\rho) \simeq -\frac{n}{e} [1 - m\rho K_1(m\rho)], \quad (2.6)$$

$$\phi(\rho) = + \left(\frac{c_2}{2c_4} \right)^{1/2} [1 + O(e^{-2\sqrt{c_2}\rho})],$$

where

$$m = e \left(\frac{c_2}{2c_4} \right)^{1/2}.$$

We are interested in exact solutions of the coupled system of Eqs. (2.5). To this end, it will be convenient to consider the energy-momentum tensor of the system:

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi^* + i e (A_\mu \phi \partial_\nu \phi^* - A_\nu \phi^* \partial_\mu \phi) + e^2 A_\mu A_\nu \phi \phi^* + F_{\mu\alpha} F^\alpha_\nu - g_{\mu\nu} \mathcal{L}.$$

For the vortex solution (2.3), the only nonvanishing components are

$$T_{\rho\rho} = \frac{1}{2\rho^2} \left(\frac{dA}{d\rho} \right)^2 + \frac{1}{2} \left(\frac{df}{d\rho} \right)^2 - \frac{f^2}{2\rho^2} (n + eA)^2 + c_2 f^2 - c_4 f^4 - \frac{c_2^2}{4c_4}, \quad (2.7)$$

$$T_{\varphi\varphi} = \frac{1}{2} \left(\frac{dA}{d\rho} \right)^2 + \frac{1}{2} f^2 (n + eA)^2 + \rho^2 \left[c_2 f^2 - c_4 f^4 - \frac{1}{2} \left(\frac{df}{d\rho} \right)^2 - \frac{c_2^2}{4c_4} \right], \quad (2.8)$$

$$T_{00} = \frac{1}{2\rho^2} \left(\frac{dA}{d\rho} \right)^2 + \frac{1}{2} \left[\left(\frac{df}{d\rho} \right)^2 + \frac{f^2}{\rho^2} (n + eA)^2 \right] - c_2 f^2 + c_4 f^4 + \frac{c_2^2}{4c_4}. \quad (2.9)$$

(We took advantage of the freedom in the definition of $T_{\mu\nu}$ in order to have $\lim_{\rho \rightarrow \infty} T_{\mu\nu} = 0$.)

The conservation law for $T_{\mu\nu}$ takes the form

$$\frac{d}{d\rho} (\rho T_{\rho\rho}) = \frac{T_{\varphi\varphi}}{\rho^2}. \quad (2.10)$$

III. DECOUPLING OF EQUATIONS OF MOTION

In this section, we shall concern ourselves with the decoupling of Eqs. (2.5). We will show that this is possible provided a particular relation between the coupling constants holds.

To begin with, we make the ansatz

$$T_{\rho\rho}(\rho) = T_{\varphi\varphi}(\rho) = 0, \quad (3.1)$$

which is consistent with the boundary conditions (2.4) and the conservation law (2.10).

From condition (3.1) and the explicit expressions of $T_{\rho\rho}$ and $T_{\varphi\varphi}$, we obtain the following equations:

$$\left(\frac{df}{d\rho} \right)^2 = \frac{f^2}{\rho^2} (n + eA)^2, \quad (3.2)$$

$$\left(\frac{1}{\rho} \frac{dA}{d\rho} \right)^2 = 2c_4 f^4 - 2c_2 f^2 + \frac{c_2^2}{2c_4}. \quad (3.3)$$

Integrating Eq. (3.2) one finds

$$f(\rho) = + \left(\frac{c_2}{2c_4} \right)^{1/2} \exp \left\{ \pm \int_\infty^\rho [n + eA(\rho')] \frac{d\rho'}{\rho'} \right\}. \quad (3.4)$$

In order to satisfy the boundary conditions (2.4), one must take the upper sign in Eq. (3.4). Doing so we get

$$\frac{df}{d\rho} = (n + eA) \frac{f}{\rho}. \quad (3.5)$$

One then obtains [Eqs. (3.3)]

$$\frac{1}{\rho} \frac{dA}{d\rho} = \pm \sqrt{2c_4} \left[f(\rho)^2 - \frac{c_2}{2c_4} \right]. \quad (3.6)$$

One can easily see that any solution of Eqs. (3.5) and (3.6) verifies the equations of motion, provided the following relation holds:

$$e^2 = 8c_4. \quad (3.7)$$

In other words, for the particular relation (3.7), Eqs. (3.5) and (3.6) are equivalent to Eqs. (2.5).

The main point now is that in this case one can decouple the system of nonlinear equations. Indeed, from Eqs. (3.5) and (3.6) one gets

$$\frac{d^2 A}{d\rho^2} - \frac{1}{\rho} \frac{dA}{d\rho} - 2(n + eA) \left(\frac{2c_2}{e} + \frac{1}{\rho} \frac{dA}{d\rho} \right) = 0, \quad (3.8)$$

$$f(\rho)^2 = \frac{4c_2}{e^2} + \frac{2}{e} \frac{1}{\rho} \frac{dA}{d\rho}. \quad (3.9)$$

We see that if relation (3.7) holds, the model depends only on two parameters: the mass m of the vector particle ($m = 2\sqrt{c_2}$)—which coincides with that of the scalar one—and the charge e .

As we pointed out before, the static solution of the Higgs-type Lagrangian (2.1) was identified with the Ginzburg-Landau theory of type-II superconductivity. When $e^2 < 8c_4$ the system presents type-II superconductivity and for $e^2 > 8c_4$ a complete Meissner effect takes place (type-I superconductivity).²

It can also be pointed out that an identical relation to (3.7) is necessary to make the Higgs model supersymmetric. However, the supersymmetric model studied (Ref. 3) includes a Dirac spinor in addition to the scalar and vector fields.

Before studying the solutions of Eqs. (3.8) and (3.9), we will find the energy of the vortex in the rest frame, that is, its mass. From Eqs. (2.9), (3.2), (3.3), and (3.9)

$$T_{00} = \frac{1}{\rho^2} \left(\frac{dA}{d\rho} \right)^2 + \frac{2}{e} \left(\frac{1}{\rho} \frac{dA}{d\rho} + \frac{2c_2}{e} \right) \left(\frac{n+eA}{\rho} \right)^2. \quad (3.10)$$

One can easily integrate the classical expression for the energy density, thus obtaining

$$\begin{aligned} M_n &= 2\pi \int_0^\infty \rho \, d\rho \, T_{00}(\rho) \\ &= \frac{2n\pi}{e} \left(-\frac{1}{\rho} \frac{dA}{d\rho} \right)_{\rho=0} \\ &= n \frac{\pi m^2}{e^2}. \end{aligned} \quad (3.11)$$

IV. SOLUTION OF EQUATIONS OF MOTION

As we pointed out in the preceding section, one can obtain solutions of the equations of motion by solving the simpler equation (3.8), provided $e^2 = 8c_4$.

For brevity of subsequent notation we define

$$\begin{aligned} \xi &\equiv \sqrt{c_2} \, \rho = \frac{1}{2} m \rho, \\ \mathcal{K}(\xi) &\equiv n + eA(\rho). \end{aligned} \quad (4.1)$$

Then Eqs. (3.8), (3.9) become

$$\frac{d^2 \mathcal{K}}{d\xi^2} - \frac{1}{\xi} \frac{d\mathcal{K}}{d\xi} - 4\mathcal{K} = 2 \frac{\mathcal{K}}{\xi} \frac{d\mathcal{K}}{d\xi}, \quad (4.2)$$

$$f(\xi)^2 = \frac{m^2}{e^2} \left(1 + \frac{1}{2\xi} \frac{d\mathcal{K}}{d\xi} \right). \quad (4.3)$$

The boundary conditions are now

$$\mathcal{K}_n(0) = n, \quad \lim_{\xi \rightarrow \infty} \mathcal{K}_n(\xi) = 0. \quad (4.4)$$

As Eq. (4.2) is invariant under the transformation $\xi \rightarrow -\xi$ [and $\mathcal{K}(0) \neq 0$] we seek an expansion in even powers of ξ of the form

$$\mathcal{K}_n(\xi) = n + \sum_{s=1}^{\infty} D_n^s \xi^{2s}. \quad (4.5)$$

Here the subscript n stands for the number of flux quanta. Equation (4.2) implies the following conditions for the expansion coefficients:

$$D_n^1 = -1; \quad (4.6)$$

if $n \geq 2$,

$$D_n^s = 0 \text{ for } 2 \leq s \leq n, \quad (4.7)$$

and the recursion relation for $s \geq n+2$

$$D_n^s = \frac{1}{s(s-n-1)} \sum_{r=n+1}^{s-1} r D_n^r D_n^{s-r}. \quad (4.8)$$

Relation (4.6) ensures that the Higgs field is zero for $\rho=0$. Moreover, Eq. (4.7) shows that this zero is of order n . Indeed, from Eqs. (4.3) and (4.5), for $\xi \rightarrow 0$,

$$f_n(\xi) = \frac{m}{e} [(n+1)D_n^{n+1}]^{1/2} \xi^n [1 + O(\xi^2)]. \quad (4.9)$$

The recursion relation (4.8) fixes all the D_n^s in terms of D_n^1 and D_n^{n+1} . We know that $D_n^1 = -1$, but do not yet know the value of D_n^{n+1} . This coefficient will be fixed by the boundary condition at infinity. We have thus obtained an expression for $\mathcal{K}(\xi)$, good for small values of ξ . We now seek an expression of the same solution valid at large distances.

In order to do that, we will write an integral equation equivalent to Eq. (4.2) with boundary conditions (4.4). Taking into account the fact that the left-hand side of Eq. (4.2) can be related to the modified Bessel equation of order one, one can obtain a nonlinear integral equation by using the well-known Green's function method

$$\begin{aligned} \mathcal{K}_n(\xi) &= n 2\xi K_1(2\xi) \\ &\quad - 2\xi K_1(2\xi) \int_0^\xi I_1(2\eta) \frac{\mathcal{K}_n(\eta)}{\eta} \frac{d\mathcal{K}_n}{d\eta} d\eta \\ &\quad - 2\xi I_1(2\xi) \int_\xi^\infty K_1(2\eta) \frac{\mathcal{K}_n(\eta)}{\eta} \frac{d\mathcal{K}_n}{d\eta} d\eta. \end{aligned} \quad (4.10)$$

It is easy to check that Eq. (4.5) gives the same solution as Eq. (4.10) for small ξ . We will use this fact in order to find the numerical value of D_n^{n+1} .

The first term on the right-hand side of Eq. (4.10) corresponds to the approximate solution given by Nielsen and Olesen for the vector potential. It is not necessary to determine D_n^{n+1} to obtain some general properties of the solution. For example, we can find the asymptotic behavior of $\mathcal{K}$ for large ξ . Integrating Eq. (4.10), one obtains

$$\begin{aligned} \mathcal{K}_n(\xi) &= Z_n 2\xi K_1(2\xi) + (Z_n)^2 16\xi \left[I_1(2\xi) \int_\xi^\infty d\eta \eta K_1(2\eta)^2 K_0(2\eta) - K_1(2\xi) \int_\xi^\infty d\eta \eta I_1(2\eta) K_1(2\eta) K_0(2\eta) \right] + O(e^{-6\xi}) \\ &= \sqrt{\pi} Z_n \sqrt{\xi} e^{-2\xi} [1 + O(\xi^{-1})] - \frac{1}{3} (\sqrt{\pi} Z_n)^2 e^{-4\xi} [1 + O(\xi^{-1})] + O(e^{-6\xi}); \end{aligned} \quad (4.11)$$

here the constant Z_n is defined by

$$Z_n \equiv n - \int_0^\infty I_1(2\eta) \frac{\mathcal{K}_n(\eta)}{\eta} \frac{d\mathcal{K}_n}{d\eta} d\eta. \quad (4.12)$$

One can also find upper and lower bounds for the solution and its derivative (which is related to the magnetic field). Integrating Eq. (4.2) one obtains

$$\ln\left(1 + \frac{1}{2\xi} \frac{d\mathcal{K}_n}{d\xi}\right) = -2 \int_{\xi}^{\infty} \mathcal{K}_n(\eta) \frac{d\eta}{\eta}. \quad (4.13)$$

This equation, together with the boundary conditions for $\mathcal{K}_n(\xi)$, allows us to obtain the following inequalities:

$$n \geq \mathcal{K}_n(\xi) > 0, \quad (4.14)$$

$$-1 \leq \frac{1}{2\xi} \frac{d\mathcal{K}_n}{d\xi} < 0. \quad (4.15)$$

In the preceding relation, the equality holds only for $\xi=0$. Thus we see that the vector potential $A(\rho)$ is a monotonically decreasing function of ρ . Because the magnetic field H_z is given by the expression

$$H_z = -\frac{m^2}{4e} \frac{1}{\xi} \frac{d\mathcal{K}_n}{d\xi}, \quad (4.16)$$

we see from relation (4.15) that

$$0 < H_z \leq \frac{m^2}{2e}. \quad (4.17)$$

Moreover, one can show from Eqs. (4.3), (4.5), and (4.16) that the derivative of H_z is negative. Hence, the magnetic field decreases monotonically from its value at the origin to zero at infinity. The preceding relations allow us to make a qualitative picture of the field behavior: the magnetic field decreasing monotonically with a characteristic length $1/m$ and the scalar field increasing with the same length from zero at the origin to its vacuum value at infinity.

One can obtain more restrictive upper and lower bounds for the functions $\mathcal{K}_n(\xi)$, the constants Z_n , and the coefficients D_n^{n+1} . We show in the Appendix that the following inequalities hold:

$$n2\xi K_1(2\xi) \leq \mathcal{K}_n(\xi) < Z_n 2\xi K_1(2\xi) \quad (4.18)$$

(we see that the approximate solution of Ref. 1 is in fact a lower bound of the exact solution),

$$Z_n > \frac{3}{2}n + \left(\frac{\pi}{3\sqrt{3}} - \frac{1}{2}\right)n^2 = 1.5n + 0.1046n^2, \quad (4.19)$$

$$1.6046 < Z_1 < 3.230, \quad (4.20)$$

$$D_n^{n+1} < \frac{e^{n2\gamma}}{n+1} \quad (4.21)$$

(here γ is the Euler-Mascheroni constant),

$$|\ln(n+1)D_n^{n+1}| < 1 + 2nZ_n K_0(2), \quad (4.22)$$

and, for the case $n=1$,

$$0.128 < D_1^2 < 1.586. \quad (4.23)$$

V. EXPLICIT ONE-QUANTUM VORTEX SOLUTION

We have obtained in Sec. IV the exact vortex solution as a power-series expansion [Eq. (4.5)]. All the coefficients in this expression can be obtained from the recursion relation (4.8) provided one knows the value of D_n^{n+1} . As we have said before, this coefficient must be fixed by the boundary condition at infinity. The aim of this section is to find the numerical value of D_n^{n+1} for the lowest-energy vortex, namely $n=1$. (It is also possible to get the corresponding solution for $n>1$ in the same way.)

From Eq. (4.13) and the boundary condition (4.4) it is easy to derive the following integral equation for $\mathcal{K}_n(\xi)$:

$$\begin{aligned} \mathcal{K}_n(\xi) = n - \xi^2 + 2 \left\{ \exp \left[2 \int_0^{\infty} \ln \eta \mathcal{K}'_n(\eta) d\eta \right] \right\} \\ \times \int_0^{\xi} d\eta \eta^{1+2\mathcal{K}_n(\eta)} \exp \left[-2 \int_0^{\eta} \ln \xi \mathcal{K}'_n(\xi) d\xi \right]. \end{aligned} \quad (5.1)$$

Expanding the right-hand side in Eq. (5.1) for small values of ξ and comparing with the expansion (4.4), it follows that

$$D_n^{n+1} = \frac{1}{n+1} \exp \left[2 \int_0^{\infty} d\xi \ln \xi \mathcal{K}'_n(\xi) \right]. \quad (5.2)$$

For the $n=1$ case, Eqs. (5.2) and (4.12) can be written as

$$\ln 2D_1^2 = 2 \int_0^{\infty} d\xi \ln \xi \frac{d\mathcal{K}_1(\xi)}{d\xi}, \quad (5.3)$$

$$Z_1 = 1 - \int_0^{\infty} I_1(2\xi) \frac{\mathcal{K}_1(\xi)}{\xi} \frac{d\mathcal{K}_1}{d\xi} d\xi. \quad (5.4)$$

Splitting the integration in these equations into two intervals $(0, b)$ and (b, ∞) and replacing the adequate expression in each one—the power series (4.5) in the first interval and the asymptotic expansion (4.11) in the second one—one arrives at a system of two coupled nonlinear equations for Z_1 and D_1^2 . In this way, one has reduced the problem of solving the system of nonlinear differential equations (2.5) to a system of nonlinear transcendental numerical equations.

We have solved these equations numerically. In doing so, we have used not only the power series (4.5) in the interval $(0, b)$ but also its analytic continuations. In this way, we have checked that the solution was independent of the value chosen for b . We have also verified the stability of the results. The values obtained for Z_1 and D_1^2 are

$$Z_1 = 1.7079 \dots, \quad (5.5)$$

$$D_1^2 = 0.72791 \dots. \quad (5.6)$$

With these values for Z_1 and D_1^2 and the recursion relation (4.8)

$$D_1^s = \frac{1}{s(s-2)} \sum_{r=2}^{s-1} r D_1^r D_1^{s-r} \quad (5.7)$$

one knows all the coefficients of the power expansion of the solution

$$\vec{A}(\vec{r}) = \frac{\hat{e}_\varphi}{e\rho} \left\{ 1 - Z_1 m K_1(m\rho) - (Z_1)^2 2m\rho \left[I_1(m\rho) \int_{m\rho}^\infty \alpha d\alpha K_1(\alpha)^2 K_0(\alpha) - K_1(m\rho) \int_{m\rho}^\infty d\alpha K_1(\alpha) I_1(\alpha) K_0(\alpha) \right] + O((Z_1)^3 e^{-3m\rho}) \right\}, \quad (5.10)$$

$$\phi(\vec{r}) = e^{-i\varphi} \frac{m}{e} \left\{ 1 - Z_1 K_0(m\rho) - \frac{(Z_1)^2}{2} K_0(m\rho)^2 + 2(Z_1)^2 \left[I_0(m\rho) \int_{m\rho}^\infty \alpha d\alpha K_1^2(\alpha) K_0(\alpha) + K_0(m\rho) \int_{m\rho}^\infty \alpha d\alpha I_1(\alpha) K_1(\alpha) K_0(\alpha) \right] + O((Z_1)^3 e^{-3m\rho}) \right\}. \quad (5.11)$$

In Fig. 1 we plot the exact solution for the modulus of the scalar field and the magnetic field. In Fig. 2 we present the plot of the vector potential.

We have investigated the convergence of the series expansion (5.11) and its analytic continuations from the numerical values of its coefficients. We found that the radius of convergence of series (5.11) is

$$R = 1.255. \quad (5.12)$$

This, together with the convergence radius of the analytic prolongations of series (5.11), shows that the solution has singularities at $\xi = \pm iR$.

VI. FORM FACTORS

We have thus obtained the exact classical solution of a vortex of one quantum of flux, provided

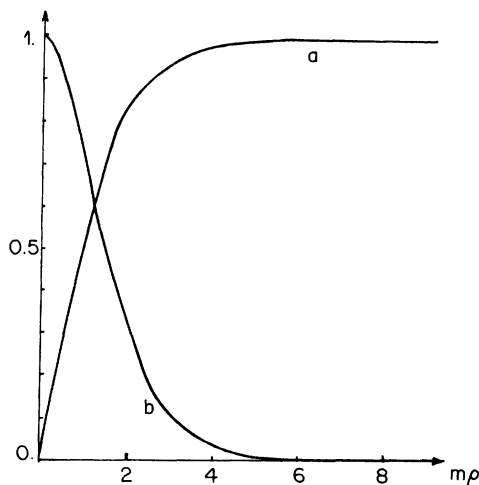


FIG. 1. The curves a and b refer to $(e/m)|\phi(\vec{r})|$ and $(2e/m^2)H_z(\rho)$, respectively, for the case $n=1$.

$$\vec{A}(\vec{r}) = \frac{\hat{e}_\varphi}{e\rho} \left[c_2 \rho^2 - \sum_{s=2}^\infty D_1^s (c_2 \rho^2)^s \right], \quad (5.8)$$

$$\phi(\vec{r}) = e^{-i\varphi} \frac{m}{e} \left[\sum_{s=2}^\infty s D_1^s (c_2 \rho^2)^{s-1} \right]^{1/2} \quad (5.9)$$

and also its asymptotic expansion from Eq. (4.11)

relation (3.7) between the coupling constants holds. It is clear that one can also get the corresponding solution for $n > 1$ by the same method as the one exposed in Sec. V. We also obtained bounds for the n -quanta solution and its parameters.

The mass spectrum of the theory at the classical level is found to be composed of a scalar and a vector particle with the same mass m and the vortices, with mass

$$M_n = n \frac{\pi m^2}{e^2}. \quad (6.1)$$

This expression can be regarded as the vortex mass per unit length. It can be pointed out that the approximate solution of Ref. 1 disagrees with this result because it gives an estimate of the energy

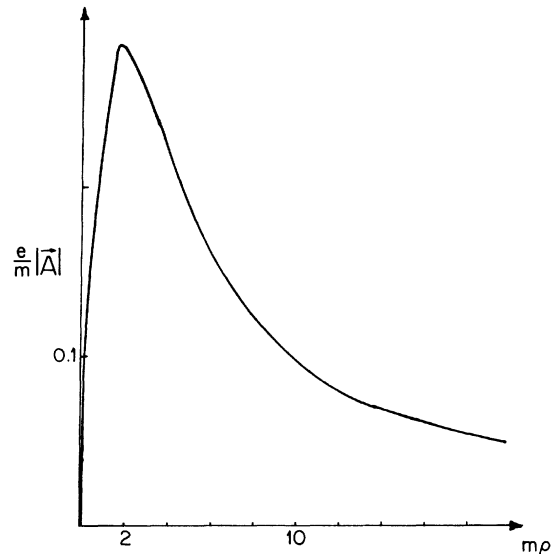


FIG. 2. A plot of $(e/m)|\vec{A}(\rho)|$ for the case $n=1$.

with increases with n^2 instead of n . We see from Eq. (6.1) that the decay of an n -quanta vortex ($n \geq 2$) in n_i -quanta vortices ($n = \sum_i n_i$) is energetically allowed. However, we do not know if there is any conservation law that makes them stable, as is the case in the sine-Gordon model.⁴

As a first application of the solution, we have calculated the scalar and vector field form factors in the classical limit.⁵ In this limit, from Eq. (2.3)

$$\begin{aligned} \langle n\vec{p} | \phi_{op}(0) | n\vec{p}' \rangle &= \int \frac{d^2x}{2\pi} e^{-i\vec{x} \cdot (\vec{p} - \vec{p}')} \phi_n(\vec{x}) \\ &= (ie^{i\alpha})^n F_n(|\vec{q}|), \end{aligned} \quad (6.2)$$

$$\begin{aligned} \langle n\vec{p} | \vec{A}_{op}(0) | n\vec{p}' \rangle &= \int \frac{d^2x}{2\pi} e^{-i\vec{x} \cdot (\vec{p} - \vec{p}')} \vec{A}_n(\vec{x}) \\ &= \frac{i}{e} \epsilon_{0\kappa i} q^\kappa \hat{e}_i G_n(|\vec{q}|), \end{aligned} \quad (6.3)$$

where

$$F_n(|\vec{q}|) = \int_0^\infty \rho J_n(|\vec{q}|\rho) f_n(\rho) d\rho, \quad (6.4)$$

$$G_n(|\vec{q}|) = \int_0^\infty [n - \mathcal{J}_n(\rho)] J_1(|\vec{q}|\rho) d\rho. \quad (6.5)$$

Here $\vec{q} = \vec{p} - \vec{p}'$, where α is its polar angle. The ket $|n\vec{p}\rangle$ corresponds to an n -quanta vortex state with momentum $\vec{p}$.

It should be remarked that the Goldstone boson has not been gauged away. One sees that the vector form factor has no longitudinal component and that the scalar one is singular at $t = -\vec{q}^2 = 0$. By Fourier-transforming the first two terms in Eq. (4.11), one gets

$$\begin{aligned} F_n(\sqrt{-t}) &= \frac{m}{e} \left[\frac{n}{t} + Z_n \frac{\Gamma(\frac{1}{2}n+1)^2}{n!} \frac{(\sqrt{-t}/m)^n}{t-m^2} F\left(\frac{n}{2}, \frac{n}{2}; n+1; \frac{t}{m^2}\right) \right. \\ &\quad \left. + \dots \right]. \end{aligned} \quad (6.6)$$

This shows explicitly the Goldstone boson pole at $t=0$ and a branch cut from $t=m^2$ to plus infinity,

$$I_1(2\eta)K_1(2\xi) - I_1(2\xi)K_1(2\eta) = \int_0^{\ln(\eta/\xi)} d\epsilon I_0(2(\xi^2 + \eta^2 - 2\xi\eta \cosh\epsilon)^{1/2}) \cosh\epsilon \geq 0 \quad \text{if } \eta \geq \xi \quad (A1)$$

and the fact that $\mathcal{K}_n(\xi)\mathcal{K}'_n(\xi) < 0$, one obtains from the integral expression (4.10)

$$n 2\xi K_1(2\xi) \leq \mathcal{K}_n(\xi) < Z_n 2\xi K_1(2\xi). \quad (A2)$$

From its definition, one can get bounds for Z_n . By partial integration of Eq. (4.12), Z_n becomes

$$Z_n = \frac{3}{2}n + \int_0^\infty \mathcal{K}(\xi)^2 I_2(2\xi) \frac{d\xi}{\xi}. \quad (A3)$$

which corresponds to the massive-scalar-massless-scalar boson production. One should avoid this problem by an appropriate gauge transformation.

We can easily calculate the vortex-meson coupling constants from the asymptotic expansion (4.11). For the vector meson, we find

$$G_{A,n} \equiv \lim_{t \rightarrow m^2} \left[\frac{t-m^2}{i} G_n(\sqrt{-t}) \right] = \frac{m}{e} Z_n, \quad (6.7)$$

and similarly for the scalar meson

$$G_{\phi,n} \equiv \lim_{t \rightarrow m^2} [(t-m^2)F_n(\sqrt{-t})] = (-1)^{n+1} \frac{m}{e} Z_n. \quad (6.8)$$

Both form factors have singularities each time $\sqrt{t}$ reaches an integer multiple of m , as can be seen from the large- ρ behavior of the solution.

For the one-vortex case, we can rewrite the scalar form factor in a covariant way,

$$\begin{aligned} \langle 1\vec{p} | \phi_{op}(0) | 1\vec{p}' \rangle &= \frac{i}{[\epsilon(\vec{p})\epsilon(\vec{p}')]^{1/2}} \epsilon_{\sigma\mu\lambda} F^\mu q^\lambda p^\sigma F_1(\sqrt{-t})/\sqrt{-t}, \end{aligned}$$

where $t = q^\mu q_\mu = (q^0)^2 - (q^1)^2 - (q^2)^2$ and F^μ is a three-vector which has the form

$$F^\mu = (0, i, 1)$$

in the frame in which $q^0 = 0$.

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APPENDIX

In this appendix we will find upper and lower bounds for the function $\mathcal{K}_n(\xi)$, the coefficient D_n^{n+1} , and the constant Z_n .

Using the integral representation⁶

Then using the lower bound of relation (A2) one gets a lower bound for Z_n ,

$$Z_n > \frac{3}{2}n + n^2 \left(\frac{\pi}{3\sqrt{3}} - \frac{1}{2} \right). \quad (A4)$$

One can easily obtain an upper bound for Z_1 . Splitting the integral in expression (4.12) and using the bounds given by Eqs. (4.14), (4.15), and (4.18), one arrives at the following inequality:

$$Z_1 < 1 + 2 \int_0^a I_1(2\eta) d\eta + 8Z_1^2 \int_a^\infty \eta d\eta K_1(2\eta) K_0(2\eta) I_1(2\eta). \quad (\text{A5})$$

In consequence, the value of Z_1 must be such that the quadratic polynomial

$$R_a(Z_1) \equiv 8(Z_1)^2 \int_a^\infty \eta d\eta K_1(2\eta) K_0(2\eta) I_0(2\eta) - Z_1 + I_0(2a)$$

is positive-definite. If a is chosen so that $R_a(Z_1)$ has two real roots, one can see that the actual value of Z_1 must be smaller than the smallest root of $R_a(Z_1)$. One obtains the best upper bound for $a \sim 0.8$. This gives

$$1.6046 < Z_1 < 3.230.$$

Here we have also used Eq. (A4).

For the coefficient D_n^{n+1} one has [Eq. (5.2)]

$$D_n^{n+1} = \frac{1}{n+1} \exp \left[2 \int_0^\infty \ln \xi \mathcal{H}'_n(\xi) d\xi \right]. \quad (\text{A6})$$

Replacing the bound for $\mathcal{H}'_n(\xi)$ given by expression (A2), one obtains

$$D_n^{n+1} < \frac{e^{2n\gamma}}{n+1}, \quad (\text{A7})$$

where γ is the Euler constant.

To obtain a lower bound for D_n^{n+1} , we split the integral in Eq. (5.2) into the following form:

$$|\ln(n+1)D_n^{n+1}| \leq 2 \int_0^1 |\ln \eta| 2\eta d\eta + 2 \int_1^\infty \ln \eta 4nZ_n \eta K_0(2\eta) d\eta. \quad (\text{A8})$$

There we have used relations (4.15) and (A2).

Performing the integration in Eq. (A8), one finally obtains

$$|\ln(n+1)D_n^{n+1}| \leq 1 + 2nZ_n K_0(2). \quad (\text{A9})$$

For the one-quantum solution, inequalities (A7) and (A9) read

$$0.128 < D_1^2 < 1.586. \quad (\text{A10})$$

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